# Basic Buck Converter

## Objective

This note examines the open-loop behavior of a DC–DC buck converter using PSIM simulation. The focus is on:

- The relationship between duty cycle and average output voltage.
- The influence of inductance and capacitance on ripple and startup transient response.

The goal is to observe fundamental converter behavior before introducing closed-loop control.

## System Model

Input voltage:50VDC
Load:5Ω
Control method: Open-loop PWM
Duty ratios evaluated:D =0.3, 0.5, 0.7
Simulations were performed in PSIM to compare voltage and inductor-current waveforms under identical hardware conditions.

## Basic Principle

Operating Modes

**When the switch is ON:**
The input voltage is applied to the inductor, and energy is stored in its magnetic field.

$$V_{L,on} = V_{in} - V_{out}$$

**When the switch is OFF:**
The inductor releases stored energy through the diode into the load, maintaining continuous current flow.

$$V_{L,off} = -V_{out}$$

The average output voltage is given by:

$$V_{out} = D\, V_{in}$$

where is the duty cycle.
**Example:** For $V_{in}$=50$V$ and $D$=0.5
$V_{out} \approx 25V$.

**Circuit Parameters/Simulation Conditions**

| PARAMETER | SYMBOL | VALUE | DESCRIPTION |
|---|---|---|---|
| INPUT VOLTAGE | (Vin) | 50V | Supply voltage |
| LOAD RESISTANCE | (R) | 5Ω | Represents load |
| INDUCTOR | (L) | 1mH | Controls current ripple |
| CAPACITOR | (C) | 100μF | Smoothens output voltage |
| SWITCHING FREQUENCY | (fs) | 5kHz | Determines ripple and loss |
| DUTY CYCLE | (D) | 0.3<br>0.5<br>0.7 | Target ratio |

These values were selected to maintain continuous conduction while allowing visible ripple and transient effects.

## Simulation Results and Analysis

### Effect of Duty Cycle on Output Voltage

The simulation results show that the average output voltage increases proportionally with the PWM duty cycle.

| Duty Cycle | Theoretical Output Voltage | Simulated Output Voltage |
|---|---|---|
| 0.3 | 15V | 15V |
| 0.5 | 25V | 25V |
| 0.7 | 35V | 35V |

The theoretical values are calculated using the ideal buck converter voltage gain equation
The simulation results are expected to be close to the theoretical values. Small differences may result from simulation conditions, measurement methods, and the characteristics of the device models used.

Steady-State Behavior

- The average output voltage scaled proportionally with duty cycle.
- D=0.3 produced approximately 15V.
- D=0.5 produced approximately 25V.
- =0.7 produced approximately 35V.

The proportional relationship between $Vout$ and D was observed in the waveform comparison (Figures 2 and 3).

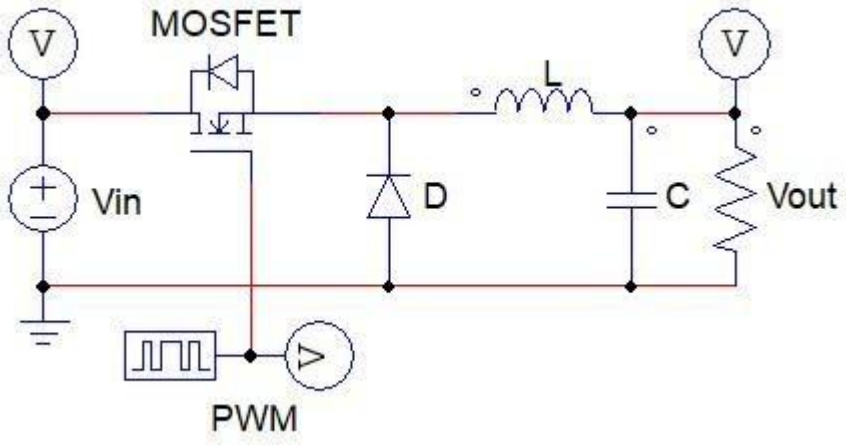



Figure 1

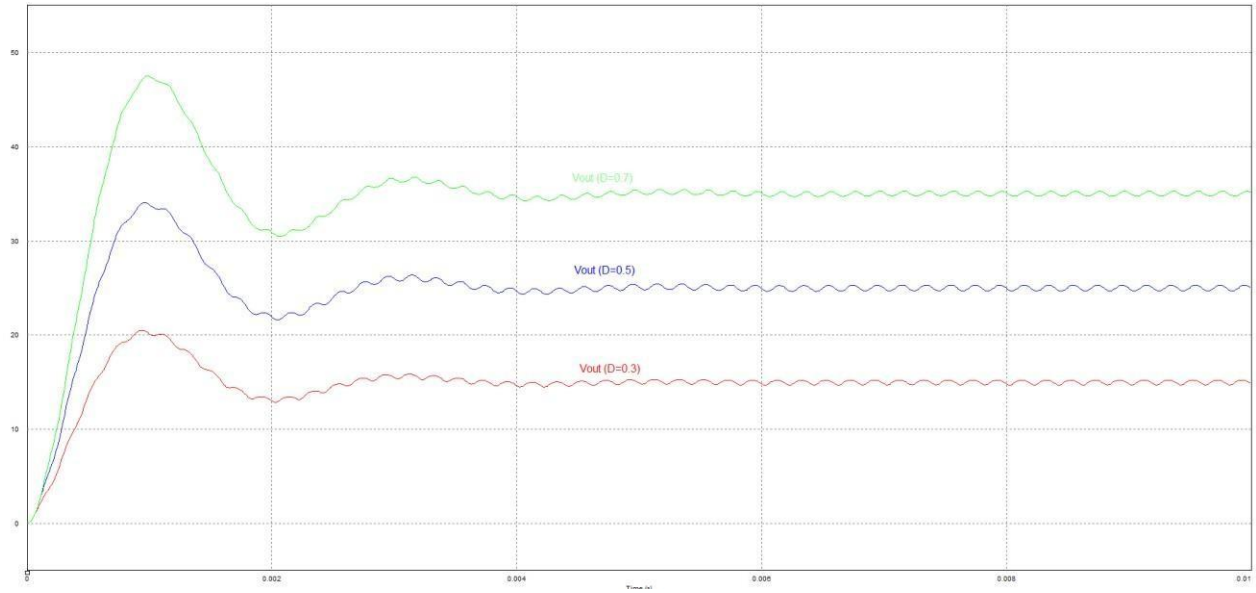



Figure 2 Comparison of output voltage waveforms at duty cycles D = 0.3, 0.5, and 0.7.

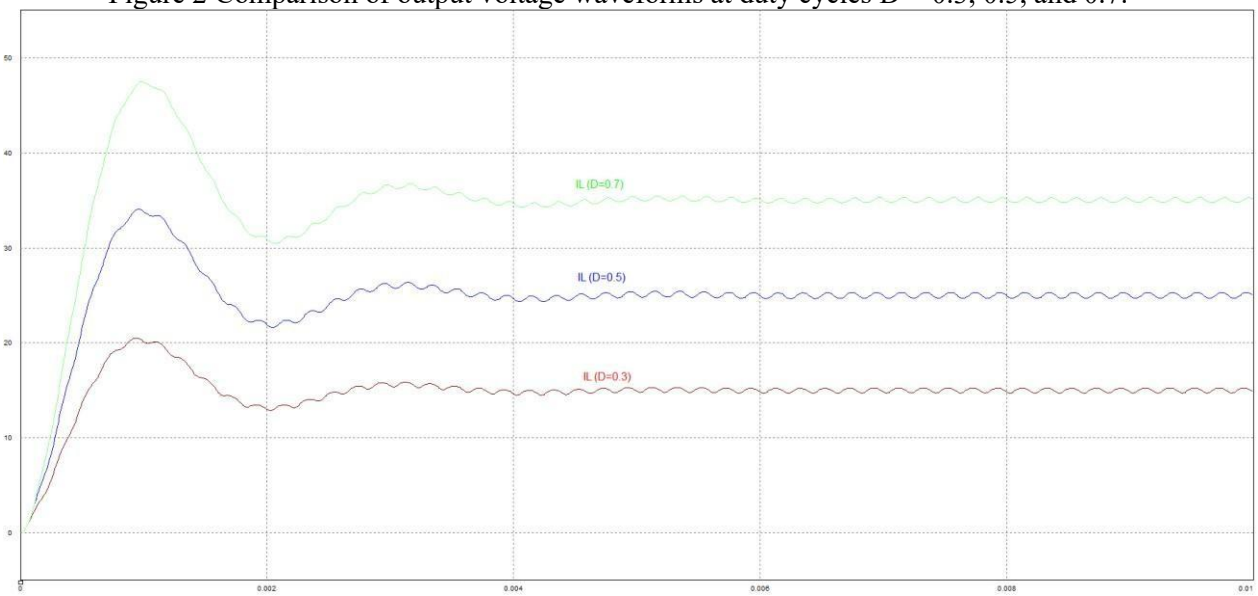



Figure 3 Comparison of inductor's current waveforms at duty cycles D = 0.3, 0.5, and 0.7.

## Startup Behavior

With L = 1 mH and C = 100 µF:

- A noticeable overshoot occurred before settling.
- Inductor current exhibited a corresponding transient peak.

When L was increased to 5 mH and C reduced to 30–50 µF:

- The voltage rise became smoother.
- Overshoot was significantly reduced.

This indicates that transient shape is strongly dependent on energy storage distribution between L and C.

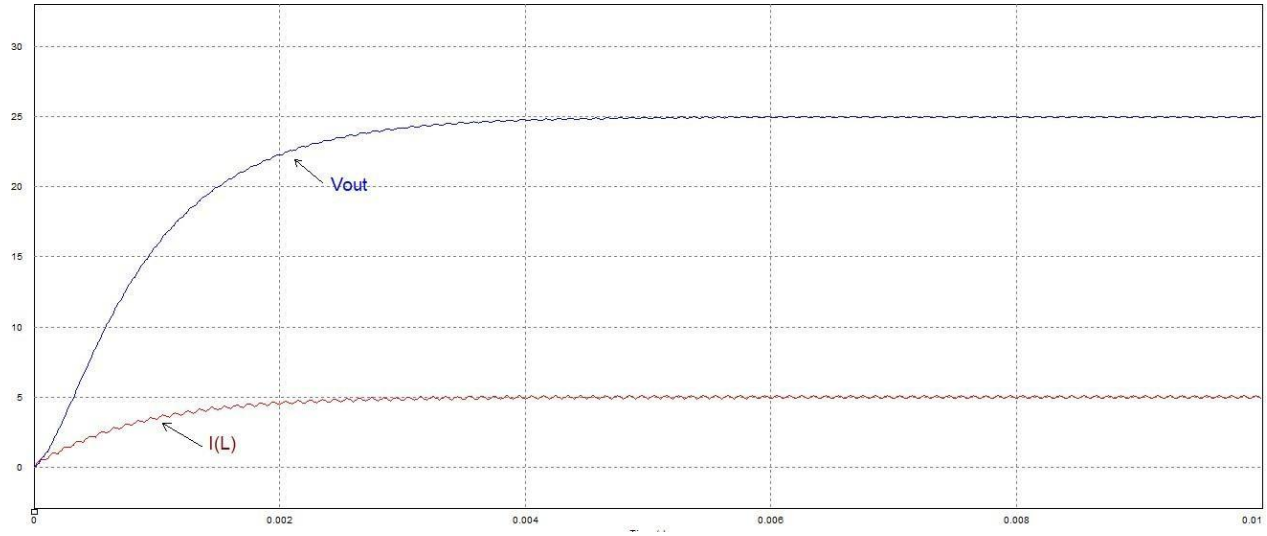



Figure 4

## Parameter Sensitivity

Duty Cycle
- Directly controls the average output voltage in open-loop operation.
- Higher D increases both output voltage and inductor current stress.

Inductance
- Larger L reduces current ripple.
- Larger L slows dynamic response.

Capacitance
- Larger C reduces voltage ripple.
- Excessive C can increase startup overshoot if inductor energy transfer is not damped.

## Summary
- In open-loop CCM operation, $Vout$ follows $Vout = D \cdot Vin$.
- Increasing duty cycle proportionally increases average output voltage.
- Startup overshoot is primarily determined by the L–C network under open-loop operation.
- Larger inductance improves damping but slows response.
- Converter transient behavior is sensitive to passive component selection even without feedback control.

This analysis establishes the open-loop behavior baseline required before implementing feedback control.